UNIVERSITÀ DI PISA

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

Candidato:
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Outline

Cuprates and Hubbard physics

Nobel Prize in Physics 1987

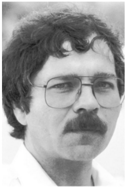
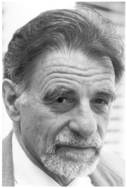


Photo from the Nobel Foundation archive.

J. Georg Bednorz

Prize share: 1/2



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K. Alexander Müller

Prize share: 1/2

Possible High T_c Superconductivity in the Ba – La – Cu – O System

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

Received April 17, 1986

What makes cuprates interesting?

- Superconductors that are insulating at normal state;
- Rich phase diagrams;
- Unclear origin of Cooper pairing;
- High-T_c superconductivity.

The timeline of high-T_c superconductivity

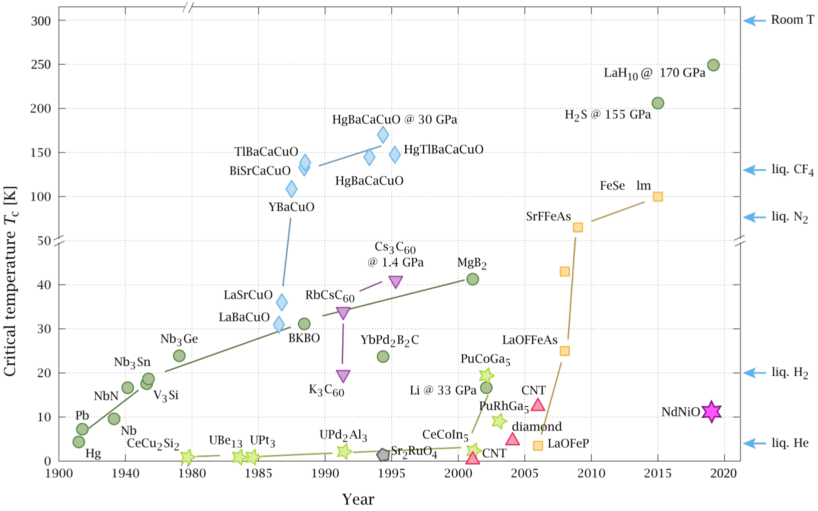


Figure 2: Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to the license CC BY-SA 4.0.

The timeline of high-T_c superconductivity

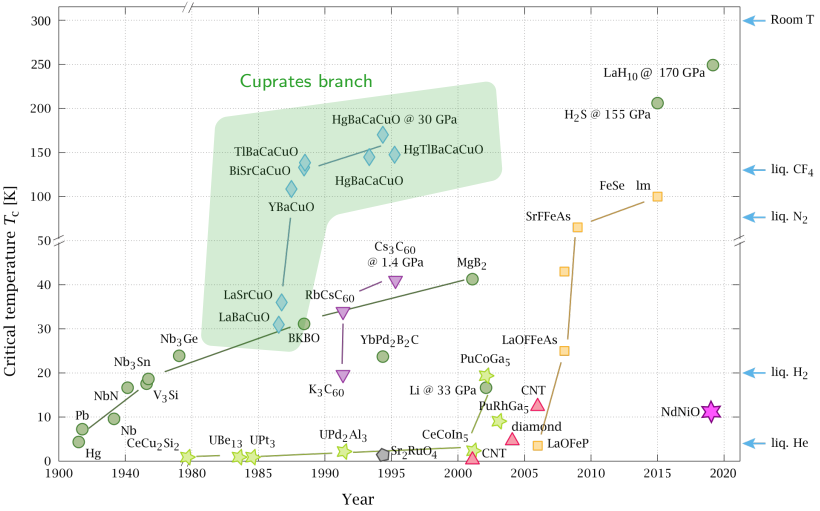


Figure 2: Timeline of the discovery of superconductivity in various compounds. The cuprates branch is represented as cyan diamonds. Image source: P. Ray and O. Gingras, distributed according to the license CC BY-SA 4.0.

The phase diagram of cuprates

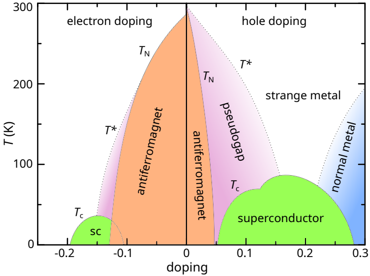
Cuprates are the prototypical example of **strongly correlated materials**:

- **Antiferromagnetism** at null and low doping;
- d-**wave superconducting** domes.

Both arise from **strong electronic correlations**.

$\Downarrow$

Failure of band theory in describing the insulation mechanism.



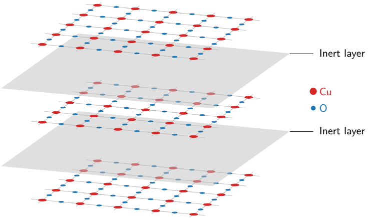
Schematic phase diagram of cuprates. Author: Holger Motzkau, Wikimedia commons, CC BY-SA 3.0.

Cuprates and the Hubbard model

These materials are stackings of:

- 2D copper oxide active layers;
- intermediate layers of inert material.

$\Downarrow$



The single-band Hubbard model on the square lattice

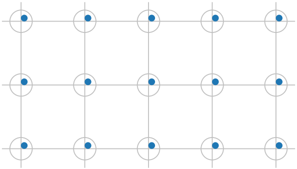
$$\hat{H}_{\mathrm{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \qquad \text{with } t, U > 0.$$

What is the origin of the insulation mechanism?

The strongly coupled Hubbard model,

$$U \gg t \,,$$

can describe **Mott insulation**.



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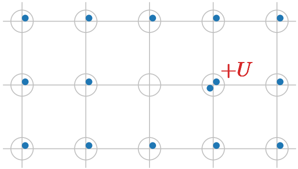

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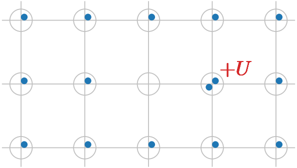
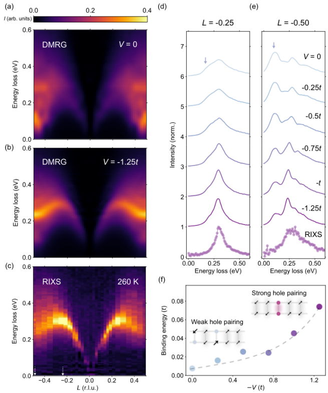
$$\Downarrow$$

Mott insulation mechanism

Electrons are **localised** as in a sort of "traffic jam".



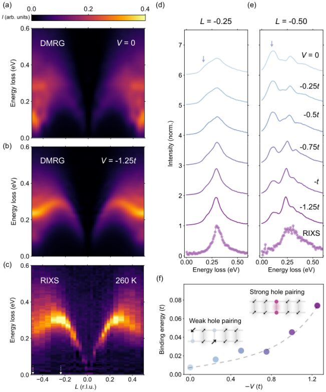
Recent developments: the extended Hubbard model



Hari Padma et al., 2025. **Physical Review X 15 (2): 021049**.

Observed: the Hubbard model does not reproduce the spectrum of magnetic excitation in certain cuprates.

Recent developments: the extended Hubbard model



Hari Padma et al., 2025. **Physical Review X 15 (2): 021049**.

Observed: the Hubbard model does not reproduce the spectrum of magnetic excitation in certain cuprates.

$$\Downarrow$$

Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\mathrm{EHM}} = \hat{H}_{\mathrm{HM}} \underbrace{- V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V} \ .$$

$\hat{H}_V$ can act **as a source of d-wave superconductivity**.

This thesis: investigating Mott physics in the EHM

The model (EHM):

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V\sum_{\langle ij\rangle}\hat{n}_{i}\hat{n}_{j}}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0.$$

The motivation: compatibility with some aspects of cuprates physics, including d-wave superconductivity.

The target: understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

The method: theoretical and numerical Hartree-Fock method, at low temperature.

The Hartree-Fock method

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0.

Variational principle:

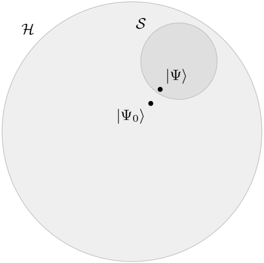
$$\forall\,|\Phi\rangle \in \mathcal{S} \subset \mathcal{H} \quad : \quad E[\Phi] \geq E_0\,.$$

$$\Downarrow$$

Constrained minimisation procedure

$$|\Psi\rangle \quad : \quad \left.\frac{\delta E[\Phi]}{\delta\,|\Phi\rangle}\right|_{|\Phi\rangle = |\Psi\rangle} = 0\,.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

The Hartree-Fock subspace of states

$\mathcal{S}$ is the set of all **gaussian states**,

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle \ , \qquad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger} \right\} = \delta_{\alpha\beta} \ ,$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

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with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

$$\Downarrow$$

Each gaussian state can be identified **uniquely** by fixing

$$\Delta^{+-}_{\alpha\beta} = \langle \hat{c}^{\dagger}_{\alpha} \hat{c}_{\beta} \rangle \,, \qquad \Delta^{++}_{\alpha\beta} = \langle \hat{c}^{\dagger}_{\alpha} \hat{c}^{\dagger}_{\beta} \rangle \,.$$

These are the **variational parameters** collected in the vector $\mathbf{v}$. We aim to find the optimal $\mathbf{v}$, such that

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle \,.$$

Symmetry breaking and HF method

Phase transition: spontaneous breaking of some symmetries of the model.

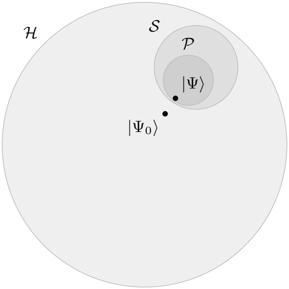
Phase X: shows symmetries Y, Z ...

$$\Downarrow$$

Restrict to $\mathcal{P} \subset \mathcal{S}$, gaussian states with symmetries Y, Z ...

$$\Downarrow$$

Eliminate from $\mathbf{v}$ the entries that do not respect the phase symmetries.



Identify symmetries $\rightarrow$ Derive $\mathbf{v}$ $\rightarrow$ Solve the variational problem.

The model (EHM):

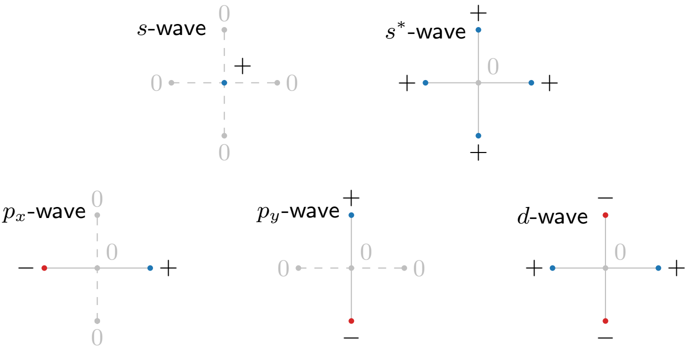
$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V\sum_{\langle ij\rangle}\hat{n}_{i}\hat{n}_{j}}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0,$$

and its symmetries:

- *Gauge invariance* (charge conservation), $\mathrm{U}(1)$.
  If broken: superconductivity.
- *Global spin rotational invariance*, $\mathrm{SU}(2)$.
  If broken: magnetism.
- *Translational invariance* in directions x and y, $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$.
  If broken: non-uniform charge and spin densities.
- *Rotational invariance* under $\pi/2$ rotations, C_4.
  If broken: nematicity.

The spatial symmetry channels

It is useful to introduce five spatial symmetry channels:



Function f on NNs: decomposition in harmonics $f^{(\gamma)}$.

The normal phase

Normal phase: no broken symmetries. Just one variational parameter,

$$\mathbf{v} = \left[u^{(s^*)} \right] \,,$$

where $u^{(s^*)}$ is the s^*-wave component of $u_{\mathbf{k}\sigma} \equiv \left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \right\rangle$.

$$\Downarrow$$

Renormalisation of kinetics

Hopping amplitude is **isotropically renormalised**,

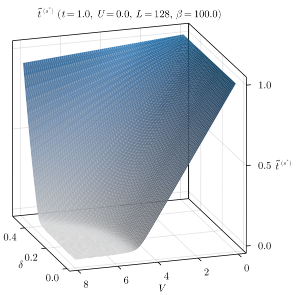
$$t \quad \rightarrow \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2} \,,$$

with the **theoretical constraint**:

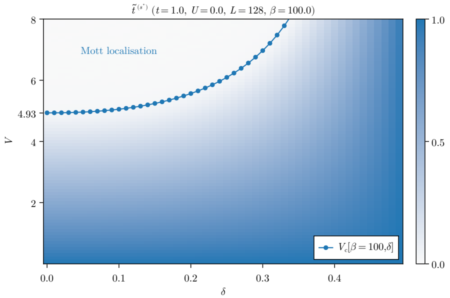
$$0 < \tilde{t}^{(s^*)} < t \qquad \Longrightarrow \qquad \text{Possible Mott localisation.}$$

Complete flattening of bare bands

In (δ, V) plane, we have determined theoretically the presence of a Mott critical boundary $V_{\mathrm{c}}[\beta, \delta]$,



(a) Numerical results for $\tilde{t}^{(s^*)}$.



(b) Critical Mott boundary.

Within the Mott plateau, electrons are completely localised and **bands are flattened**.

What we found: V-induced Mott physics

The original results

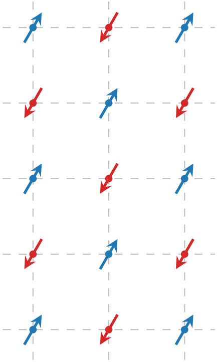
- Mott localisation induced by $-V$;
- Analytical phase boundary for the Mott transition in the (δ, V) plane.

Remarks:

- Our solution is independent of U, which is generally the source of Mott localisation.
- At Hartree-Fock level, the Hubbard model ($V = 0$) cannot describe Mott physics.

The antiferromagnetic phase

Commensurate antiferromagnetism in cuprates



In cuprates, the AF phase is **commensurate**:

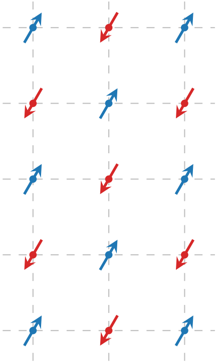
- 2-sites periodic in each direction;
- Sublattices oppositely magnetised;
- Zero net magnetisation.

$$\Downarrow$$

Staggered magnetisation m:

$$2m = |\langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle| \, ,$$

with i a site.



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$$\Downarrow$$

Staggered magnetisation m:

$$2m = \left| \langle \hat{n}_{i\uparrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle \right| ,$$

with i a site.

Half-filling restriction

We limit to $\delta = 0$ (half-filling) due to **stability considerations**.

The variational problem for the AF phase

Variational parameters for the AF phase:

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} ,$$

with $u^{(s^*)}$ defined as for the normal phase.

Renormalisation of kinetics

Mott localisation effect **formally** identical to the normal phase:

$$t \quad \rightarrow \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2} ,$$

with the **theoretical constraint**: $0 < \tilde{t}^{(s^*)} < t$.

The cooperation on magnetisation

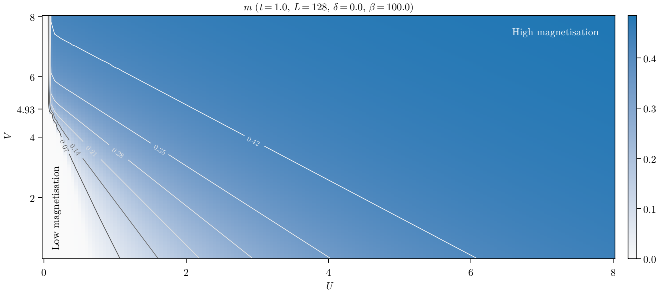


Figure 5: Numerical results for m. The gray curves indicate isovalued lines.

The cooperation on magnetisation

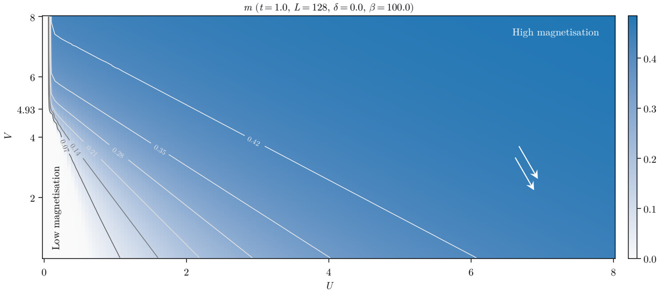


Figure 5: Numerical results for m. The gray curves indicate isovalued lines.

$$U \gg V, t$$

"Conventional" Hubbard magnetisation, induced by the ratio U/t.

The cooperation on magnetisation

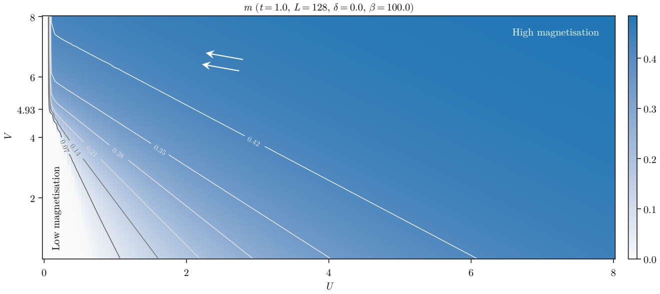


Figure 5: Numerical results for m. The gray curves indicate isovalued lines.

$V \gg U, t$

New "unconventional" source of magnetisation, induced by the ratio V/t.

The Mott enhancement of magnetisation

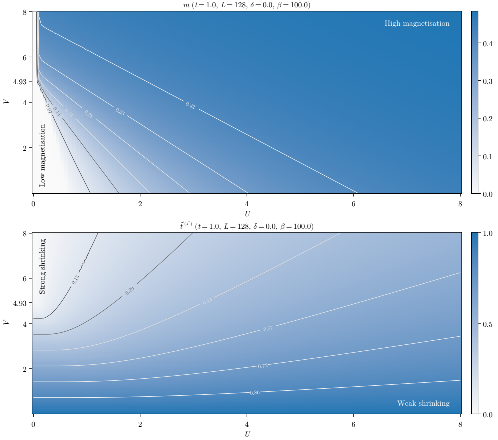


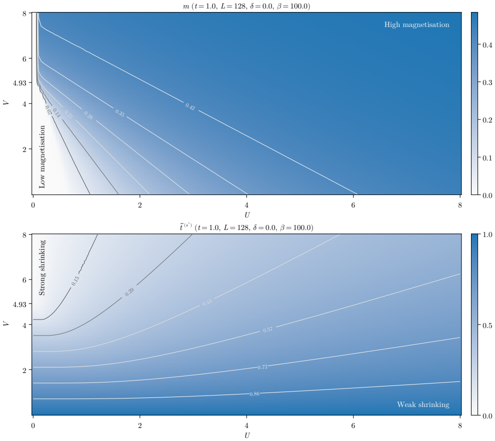
Figure 6: Numerical results for m (top) and $\tilde{t}^{(s^*)}$ (bottom). The gray curves indicate isovalued lines.

At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}\left(r^2\right) ,$$

where $r = \tilde{t}^{(s^*)}/U$.

The Mott enhancement of magnetisation



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At low temperature:

$$m \simeq \frac{1}{2} + \mathcal{O}\left(r^2\right) ,$$

where $r = \tilde{t}^{(s^*)}/U$.

$$\Downarrow$$

To enhance m, either:

- Increase U;
- **Decrease** $\tilde{t}^{(s^*)}$.

Mott localisation in the AF phase

Effective hopping is:

- reduced by V;
- **increased** by U.

$\Downarrow$

The two interactions oppose each other.

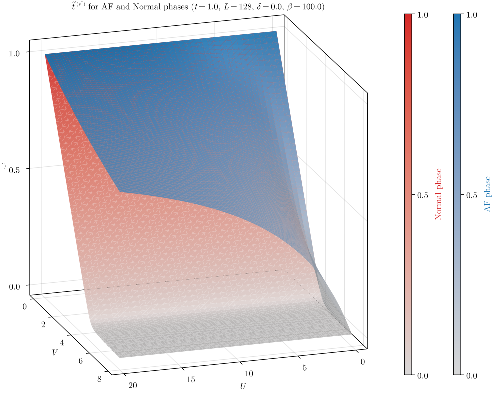


Figure 7: Comparative plot of $\tilde{t}^{(s^*)}$ for normal and AF phase.

What we found: cooperative magnetisation

The original results

- Mott localisation induced by $-V$;
- Mott localisation **enhances magnetisation**,

$$0 < \tilde{t}^{(s^*)} < t \quad \Longrightarrow \quad U/\tilde{t}^{(s^*)} > U/t\,.$$

Remarks:

- U opposes Mott localisation: Hartree-Fock artifact?

The superconducting phase

The translationally invariant SC phase

SC phase: breaking of charge conservation $\mathrm{U}(1)$.

Cooper pairing $\implies$ SC gap, roughly the "binding energy" of the pairs.

Gap symmetries and Cooper pairing channels

The gap is a function of momentum with harmonics $\tilde{\Delta}^{(\gamma)}$.

- **Singlet channel**: symmetric components,

$$\tilde{\Delta}^{(s)}, \quad \tilde{\Delta}^{(s^*)} \quad \text{and} \quad \tilde{\Delta}^{(d)}.$$

- **Triplet channel**: antisymmetric components,

$$\tilde{\Delta}^{(p_x)} \quad \text{and} \quad \tilde{\Delta}^{(p_y)}.$$

The variational problem for the singlet SC phase

We limit to the **singlet channel**. Let the quantities:

$$u_{\mathbf{k}} \equiv \left\langle \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \right\rangle , \quad w_{\mathbf{k}} \equiv \left\langle \hat{c}^{\dagger}_{\mathbf{k}\uparrow} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \right\rangle .$$

Variational parameters: symmetric harmonics,

$$\mathbf{v} = \left(u^{(s^*)}, u^{(d)}, w^{(s)}, w^{(s^*)}, w^{(d)} \right)$$

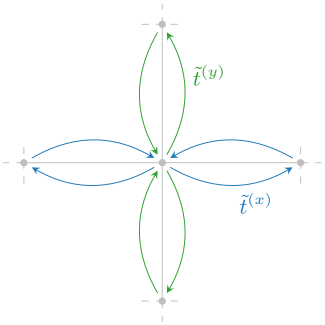
The role of each parameter

- $u^{(s^*)}$, $u^{(d)}$: renormalisation of kinetics;
- $w^{(s)}$, $w^{(s^*)}$ and $w^{(d)}$: components of the SC gap.

The nematic order parameter

In the SC phase, the renormalisation of t gives two components,

$$t \quad \to \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2}\,.$$



Anisotropic hopping

Direction-resolved effective hopping:

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2}\,,$$
$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2}\,.$$

$$\Downarrow$$

$\tilde{t}^{(d)} \neq 0 \implies$ **nematic SSB**.

Gap symmetries in the singlet channel

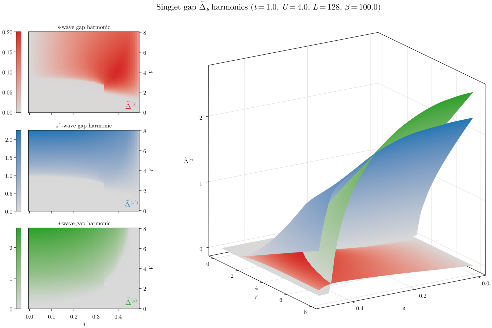


Figure 8: Singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U/t = 4$.

Gap components in **singlet** channel:

$$\tilde{\Delta}^{(s)} = -Uw^{(s)},$$
$$\tilde{\Delta}^{(s^*)} = Vw^{(s^*)},$$
$$\tilde{\Delta}^{(d)} = Vw^{(d)}.$$

Internal constraint among singlet harmonics

$$V \sum_{\gamma \neq s} \left|w^{(\gamma)}\right|^2 \geq U \left|w^{(s)}\right|^2.$$

Symmetry mixing and insurgence of nematicity

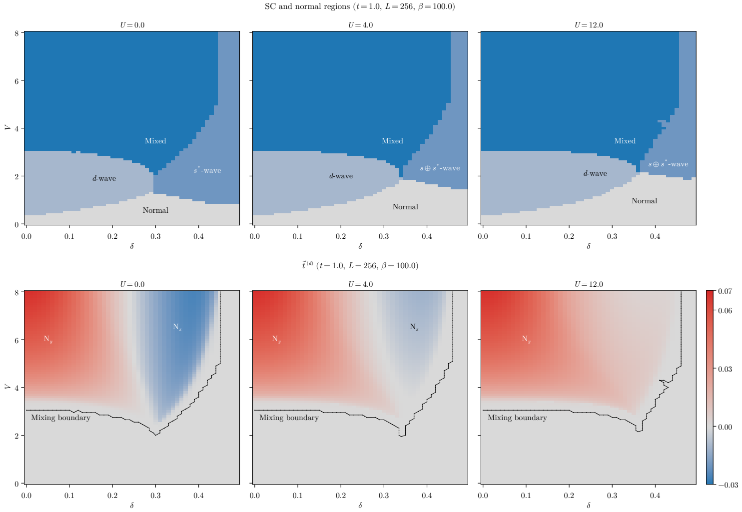


Figure 9: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.

Mott localisation in the SC phase

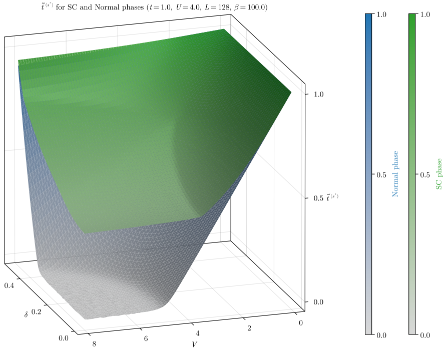


Figure 10: Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the SC phase.

Isotropic hopping renormalisation:

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} .$$

Formally identical to normal and AF phases.

Isotropic flattened component $\gg$ Anisotropic correction.

The original results

- Mott localisation induced by $-V$;
- The renormalisation of hopping is anisotropic;
- Nematic superconductivity emerges from symmetry mixing.

Remarks:

- As U grows, d-wave pairing is favoured.
- Two "flavours" of nematicity.

Conclusions and outlook

Our original results, summarised

The target: understand how the new interaction relates to Mott physics, antiferromagnetism and superconductivity.

At HF level, the non-local attraction:

- Acts as a source of Mott localisation;
- Enhances magnetisation at half filling;
- Can produce mixed-symmetry superconductivity;
- Can produce nematic superconductivity.

Further developments

From here, we can follow various paths:

1. **Same model, other phases**.
   Triplet superconductivity, breaking of translational invariance, nematic antiferromagnetism...
2. **Expand the model**.
   Second nearest neighbours hopping, more refined potentials...
3. **More advanced numerical techniques**.
   DMRG for chains and ladders; hybrid DMFT (local sector) and HF (non-local sector) to retrieve Mott physics induced by U...

or any linear combination of the three!

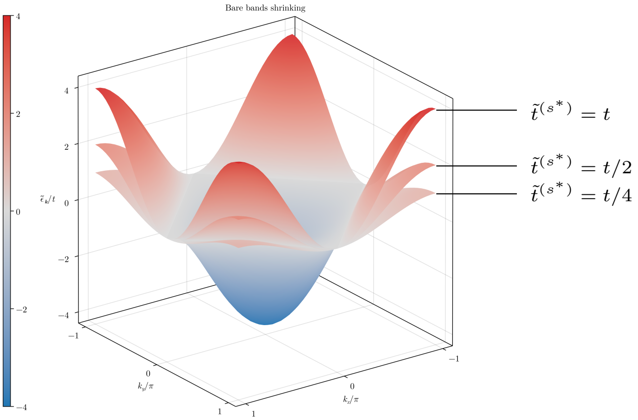
That’s all, thank you!

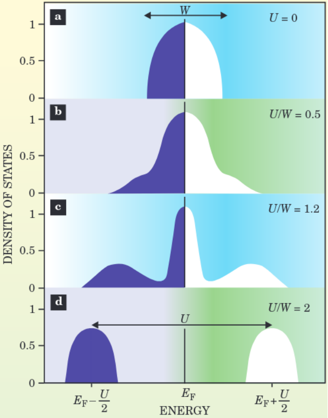
Backup 1: details on Mott localisation

Bands flattening: a signature of Mott localisation (1/2)

Renormalisation of electronic bands, $\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma}$:

$$0 < \tilde{t}^{(s^*)} < t \quad \Longrightarrow \quad \textbf{Flattening} \text{ of bands} = \text{Mott localisation.}$$





Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

As we increase the ratio U/t, the many-body DoS undergoes a transition:

(a) Free model, conventional DoS;

(b) Weak coupling, large coherent peak;

(c) Intermediate coupling, **narrow coherent peak**;

(d) Strong coupling, lower and upper Hubbard bands.

$$\Downarrow$$

Localisation = flattening of bare bands.

Mott physics, antiferromagnetism and d-wave superconductivity

Mott localisation: "electrons avoid being on the same site". How does this relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Mott localisation at half-filling.

$$\Downarrow$$

Singlet pairing $\leftrightarrow$ AF ordering.

$$\Downarrow$$

HM at strong coupling:

$$\hat{H}_{\mathrm{HM}} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E \,.$$

Superconductivity

Condensation of Cooper pairs with d-wave spatial structure.

$$\Downarrow$$

Suppressed on-site paring.

Backup 2: details on the HF method

The Hartree-Fock method in a nutshell

Hartree-Fock method: $\mathcal{S}$ is the set of all **gaussian states**, i.e. states that are non-interacting for some fermionic operators

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} |\Omega_{\gamma}\rangle \;, \qquad \left\{\hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger}\right\} = \delta_{\alpha\beta} \;,$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

$$\Downarrow$$

Wick's theorem: averages of quartic operators decompose nicely,

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}} \,.$$

The EHM hamiltonian is a quartic operator!

The self-consistency mapping

Each $|\Psi\rangle \in \mathcal{S}$ is uniquely defined by a particular choice of the **variational parameters** (HFPs)

$$\Delta^{+-}_{\alpha\beta} = \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \,, \qquad \Delta^{++}_{\alpha\beta} = \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \,.$$

Let $\mathbf{v}$ be the vector that collects all HFPs. We aim to find

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle \;,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left.\frac{\delta E[\Phi]}{\delta\,|\Phi\rangle}\right|_{|\Phi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \,.$$

Self-consistency picture

A self-consistency problem is satisfied,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v} \,,$$

being A a non-linear map **uniquely determined**.

The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map A,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}\,.$$

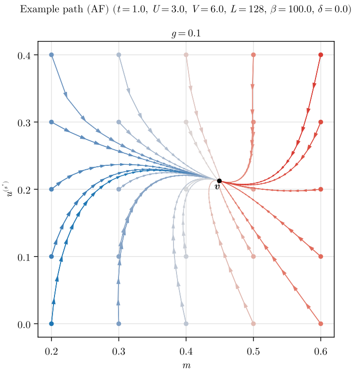
$$\Downarrow$$

Strategy: "follow" the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i\,.$$

with $i > 0$, $g \in [0, 1]$.

$$\Downarrow$$



Example path for the AF phase.

Convergence criterion:

$$\forall j \quad : \quad \left[\mathbf{v}_{i+1} - \mathbf{v}_i\right]_j < \Delta v_j\,,$$

with $\Delta\mathbf{v}$ the vector of tolerances.